TECNOLÓGICO DE ESTUDIOS SUPERIORES DE VILLA GUERRERO.

INGENIERÍA EN SISTEMAS COMPUTACIONALES

UNIDAD DE APRENDIZAJE:

SISTEMAS OPERATIVOS

EXPLORANDO EL HARDWARE DE TU SISTEMA CON LSPCI

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DOCENTE:

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Introducción

El siguiente documento fue hecho con fines educativos como evidencia de las actividades de la materia de Sistemas Operativos de la carrera de Ingeniería en Sistemas Computacionales del Tecnológico de Estudios Superiores de Villa Guerrero del presente año.

En el siguiente documento expresare el objetivo que tiene la realización de esta práctica, los requisitos mínimos de software que requiere, describiré paso a paso como ejecutar el comando lspci en la terminal de Linux, acompañado de una imagen representativa con sus respectivas descripciones y notas de imagen, por último, se expresara una conclusión donde se expresaran los conocimientos adquiridos durante la práctica.

Cabe destacar que la réplica o uso de este documento por parte de terceros recae sobre su responsabilidad y en caso de ser menores de edad se le recomienda tener y estar acompañado de sus padres durante el proceso.

Objetivo:

Familiarizarse con el comando lspci para identificar y entender los dispositivos de hardware (periféricos) conectados a tu sistema a través del bus PCI (peripheral Component Interconnect)

Requisitos

- Sistema operativo Linux
- acceso a la terminal

Procedimiento

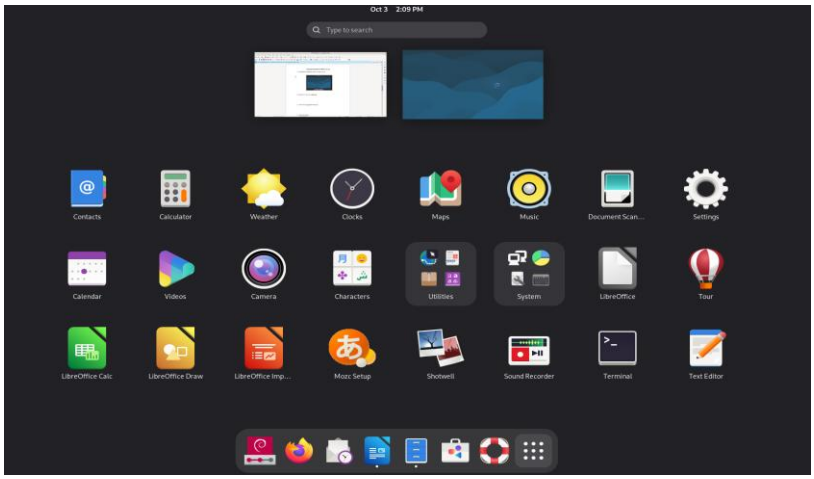
1. Abre el ordenador con el sistema operativo Linux

Ilustración 1 Escritorio de Linux



2. Presiona la tecla de Windows
3. Selecciona la opción de terminal

Ilustración 2 Menú de aplicaciones



Nota: en caso de que la tecla de Windows no despliegue la lista da clic en mosaico de aplicaciones

4. Utiliza el modo usuario sin privilegios
5. Ingresa el comando lspci en la terminal de Linux

Ilustración 3 Ejecución lspci

```
user@debian:~$ lspci
00:00.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Root Complex (rev 02)
00:00.2 IOMMU: Advanced Micro Devices, Inc. [AMD] Family 17h-19h IOMMU
00:01.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Dummy Host Bridge (rev 02)
00:02.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Dummy Host Bridge (rev 02)
00:02.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe GPP Bridge (rev 02)
00:02.3 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe GPP Bridge (rev 02)
00:02.4 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe GPP Bridge (rev 02)
00:08.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Dummy Host Bridge (rev 02)
00:08.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h Internal PCIe GPP Bridge
00:08.3 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h Internal PCIe GPP Bridge
00:14.0 SMBus: Advanced Micro Devices, Inc. [AMD] FCH SMBus Controller (rev 71)
00:14.3 ISA bridge: Advanced Micro Devices, Inc. [AMD] FCH LPC Bridge (rev 51)
00:18.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 1724
00:18.1 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 1725
00:18.2 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 1726
00:18.3 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 1727
00:18.4 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 1728
00:18.5 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 1729
00:18.6 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 172a
00:18.7 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 172b
01:00.0 Non-Volatile memory controller: KIOXIA Corporation NVMe SSD Controller BG5 (DRAM-less)
02:00.0 Network controller: Realtek Semiconductor Co., Ltd. RTL8852BE PCIe 802.11ax Wireless Network Controller
03:00.0 SD Host controller: O2 Micro, Inc. SD/MMC Card Reader Controller (rev 01)
04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Mendocino (rev c2)
04:00.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Radeon High Definition Audio Controller [Rembrandt/Strix]
04:00.2 Encryption controller: Advanced Micro Devices, Inc. [AMD] Family 19h PSP/CCP
04:00.3 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1503
04:00.4 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1504
04:00.5 Multimedia controller: Advanced Micro Devices, Inc. [AMD] Audio Coprocessor (rev 6f)
04:00.6 Audio device: Advanced Micro Devices, Inc. [AMD] Family 17h/19h/1ah HD Audio Controller
05:00.0 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1505
user@debian:~$
```

6. Una vez hayas observado los resultados anteriores
7. Ingresa el comando lspci -v y responde las siguientes preguntas:

¿Qué driver está usando tu tarjeta de red Wi-Fi (Wireless)?

Mi tarjeta es Network controller: Realtek Semiconductor Co., Ltd RTL8852BE PCIe 802.llax Wireless network controller

```
00:18.7 Host bridge: Advanced Micro Devices, Inc. [AMD] Device 172b
01:00.0 Non-Volatile memory controller: KIOXIA Corporation NVMe SSD Controller BG5 (DRAM-less)
02:00.0 Network controller: Realtek Semiconductor Co., Ltd. RTL8852BE PCIe 802.11ax Wireless Network Controller
03:00.0 SD Host controller: O2 Micro, Inc. SD/MMC Card Reader Controller (rev 01)
04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Mendocino (rev c2)
04:00.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Radeon High Definition Audio Controller [Rembrandt/Strix]
04:00.2 Encryption controller: Advanced Micro Devices, Inc. [AMD] Family 19h PSP/CCP
04:00.3 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1503
04:00.4 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1504
04:00.5 Multimedia controller: Advanced Micro Devices, Inc. [AMD] Audio Coprocessor (rev 6f)
```

Nota: Tarjeta Wi-fi del equipo donde se realizó la practica

¿Qué driver está usando tu tarjeta gráfica?

VGA controller: Advanced Micro Devices, Inc. [AMD/ATI] Mendocino (rev c2)

```
03:00.0 SD Host controller: O2 Micro, Inc. SD/MMC Card Reader Controller (rev 01)
04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Mendocino (rev c2)
04:00.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Radeon High Definition Audio Controller [Rembrandt/Strix]
04:00.2 Encryption controller: Advanced Micro Devices, Inc. [AMD] Family 19h PSP/CCP
04:00.3 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1503
04:00.4 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1504
04:00.5 Multimedia controller: Advanced Micro Devices, Inc. [AMD] Audio Coprocessor (rev 6f)
04:00.6 Audio device: Advanced Micro Devices, Inc. [AMD] Family 17h/19h/1ah HD Audio Controller
05:00.0 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1505
user@debian:~$ lspci -k
```

Nota: Tarjeta gráfica del equipo donde se realizó la practica

Ilustración 4 Ejecución de comando lspci -v

```
user@debian:~$ lspci -v
00:00.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Root Complex (rev 02)
        Subsystem: Lenovo Device 3813
        Flags: fast devsel

00:00.2 IOMMU: Advanced Micro Devices, Inc. [AMD] Family 17h-19h IOMMU
        Subsystem: Lenovo Device 3811
        Flags: bus master, fast devsel, latency 0, IRQ 25
        Capabilities: <access denied>

00:01.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Dummy Host Bridge (rev 02)
        Flags: fast devsel, IOMMU group 0

00:02.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Dummy Host Bridge (rev 02)
        Flags: fast devsel, IOMMU group 1

00:02.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe GPP Bridge (rev 02) (prog-if 00 [Normal decode])
        Subsystem: Advanced Micro Devices, Inc. [AMD] Device 1453
        Flags: bus master, fast devsel, latency 0, IRQ 33, IOMMU group 2
        Bus: primary=00, secondary=01, subordinate=01, sec-latency=0
        I/O behind bridge: [disabled] [32-bit]
        Memory behind bridge: c0700000-c07fffff [size=1M] [32-bit]
        Prefetchable memory behind bridge: [disabled] [64-bit]
        Capabilities: <access denied>
        Kernel driver in use: pcieport

00:02.3 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe GPP Bridge (rev 02) (prog-if 00 [Normal decode])
        Subsystem: Advanced Micro Devices, Inc. [AMD] Device 1453
        Flags: bus master, fast devsel, latency 0, IRQ 34, IOMMU group 3
        Bus: primary=00, secondary=02, subordinate=02, sec-latency=0
        I/O behind bridge: 2000-2fff [size=4K] [16-bit]
        Memory behind bridge: c0600000-c06fffff [size=1M] [32-bit]
        Prefetchable memory behind bridge: [disabled] [64-bit]
        Capabilities: <access denied>
        Kernel driver in use: pcieport

00:02.4 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe GPP Bridge (rev 02) (prog-if 00 [Normal decode])
        Subsystem: Advanced Micro Devices, Inc. [AMD] Device 1453
        Flags: bus master, fast devsel, latency 0, IRQ 35, IOMMU group 4
        Bus: primary=00, secondary=03, subordinate=03, sec-latency=0
        I/O behind bridge: [disabled] [32-bit]
        Memory behind bridge: c0500000-c05fffff [size=1M] [32-bit]
        Prefetchable memory behind bridge: [disabled] [64-bit]
        Capabilities: <access denied>
        Kernel driver in use: pcieport

00:08.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h PCIe Dummy Host Bridge (rev 02)
        Flags: fast devsel, IOMMU group 5

00:08.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Family 17h-19h Internal PCIe GPP Bridge (prog-if 00 [Normal decode])
        Subsystem: Advanced Micro Devices, Inc. [AMD] Family 17h-19h Internal PCIe GPP Bridge
        Flags: bus master, fast devsel, latency 0, IRQ 36, IOMMU group 6
        Bus: primary=00, secondary=04, subordinate=04, sec-latency=0
        I/O behind bridge: 1000-1fff [size=4K] [16-bit]
        Memory behind bridge: c0100000-c04fffff [size=4M] [32-bit]
```

```
        Memory at c0700000 (64-bit, non-prefetchable) [size=16K]
        Capabilities: <access denied>
        Kernel driver in use: nvme
        Kernel modules: nvme

02:00.0 Network controller: Realtek Semiconductor Co., Ltd. RTL8852BE PCIe 802.11ax Wireless Network Controller
        Subsystem: Lenovo Device 4853
        Flags: bus master, fast devsel, latency 0, IRQ 85, IOMMU group 11
        I/O ports at 2000 [size=256]
        Memory at c0600000 (64-bit, non-prefetchable) [size=1M]
        Capabilities: <access denied>
        Kernel driver in use: rtw89_8852be
        Kernel modules: rtw89_8852be

03:00.0 SD Host controller: O2 Micro, Inc. SD/MMC Card Reader Controller (rev 01) (prog-if 01)
        DeviceName: Realtek
        Subsystem: O2 Micro, Inc. Device 0002
        Flags: bus master, fast devsel, latency 0, IRQ 41, IOMMU group 12
        Memory at c0501000 (32-bit, non-prefetchable) [size=4K]
        Memory at c0500000 (32-bit, non-prefetchable) [size=2K]
        Capabilities: <access denied>
        Kernel driver in use: sdhci-pci
        Kernel modules: sdhci_pci

04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Mendocino (rev c2) (prog-if 00 [VGA controller])
        Subsystem: Lenovo Device 380e
        Flags: bus master, fast devsel, latency 0, IRQ 61, IOMMU group 13
        Memory at 7ce0000000 (64-bit, prefetchable) [size=256M]
        Memory at 7cf0000000 (64-bit, prefetchable) [size=2M]
        I/O ports at 1000 [size=256]
        Memory at c0400000 (32-bit, non-prefetchable) [size=512K]
        Capabilities: <access denied>
        Kernel driver in use: amdgpu
        Kernel modules: amdgpu

04:00.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Radeon High Definition Audio Controller [Rembrandt/Strix] (prog-if 00 [HDA compatible])
        Subsystem: Lenovo Device 3812
        Flags: bus master, fast devsel, latency 0, IRQ 83, IOMMU group 14
        Memory at c04c8000 (32-bit, non-prefetchable) [size=16K]
        Capabilities: <access denied>
        Kernel driver in use: snd_hda_intel
        Kernel modules: snd_hda_intel

04:00.2 Encryption controller: Advanced Micro Devices, Inc. [AMD] Family 19h PSP/CCP
        Subsystem: Lenovo Device 382c
        Flags: bus master, fast devsel, latency 0, IRQ 24, IOMMU group 15
        Memory at c0300000 (32-bit, non-prefetchable) [size=1M]
        Memory at c04cc000 (32-bit, non-prefetchable) [size=8K]
        Capabilities: <access denied>
        Kernel driver in use: ccp
        Kernel modules: ccp

04:00.3 USB controller: Advanced Micro Devices, Inc. [AMD] Device 1503 (prog-if 30 [XHCI])
        Subsystem: Lenovo Device 3809
        Flags: bus master, fast devsel, latency 0, IRQ 52, IOMMU group 16
```

Nota: Continuación de los resultados del comando lspci -v

Conclusión:

El comando lspci muestra todos los dispositivos conectados al bus PCI del sistema.

Permite identificar tarjetas de red, tarjetas gráficas, controladores USB, audio y otros dispositivos de hardware, con una descripción completa muy útil cuando requieres cambiar algún componente con alguna compatibilidad cercana.

Comando lspci -v muestra información detallada de cada dispositivo PCI, Incluye recursos utilizados, módulos y características del hardware.

Importancia del comando lspci

- Ayuda a identificar el hardware instalado.
- Permite verificar que los dispositivos sean reconocidos por Linux.
- Facilita la identificación de los controladores (drivers) utilizados por cada dispositivo.

En lo personal es una función describe detalladamente todos los componentes de mi computadora, cuyos componentes yo no podría visualizar sus especificaciones al menos sin desarmar por completo mi computadora, lo cual me parece mas especifico que en las especificaciones que me muestra Windows en su sección de acerca del equipo.